

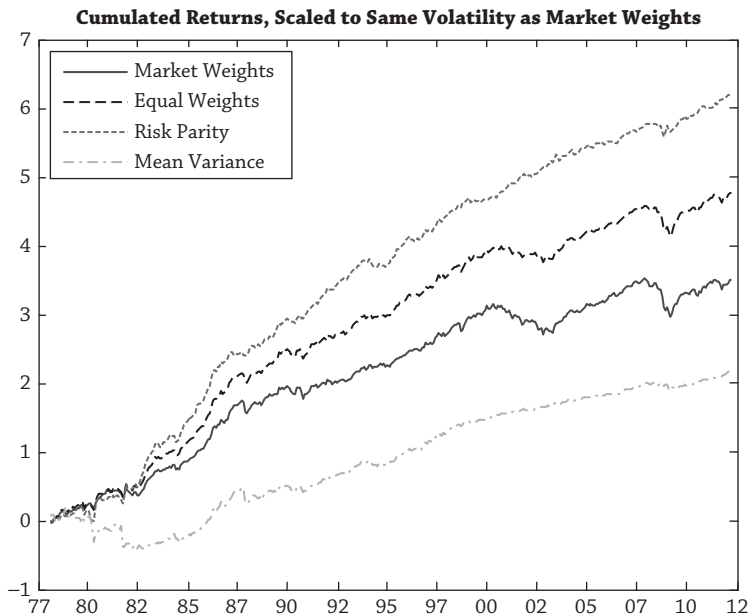


Figure 3.15

than high volatility assets, which I cover in chapter 10, and the minimum variance portfolio overweights low volatility stocks. The last two strategies in Table 3.14 are the Kelly Rule and the Proportional to Sharpe Ratio strategies. Both also outperform the Mean-Variance Weights and the market portfolio in terms of Sharpe ratios. You would have been better off, however, using the simple 1/*N* strategies in both cases.

Figure 3.15 plots cumulated returns of the Market Weights, Equal Weights, Risk Parity (Variance), and Mean-Variance strategies. All these returns are scaled to have the same volatility as the passive market weight strategy. The dominance of the Equal Weights and Risk Parity strategies is obvious. In addition, Figure 3.15 shows that the Risk Parity strategy has the smallest drawdown movements of the four strategies.

5.2. WHY DOES UNRESTRICTED MEAN-VARIANCE PERFORM SO BADLY?

The optimal mean-variance portfolio is a complex function of estimated means, volatilities, and correlations of asset returns. There are many parameters to estimate. Optimized mean-variance portfolios can blow up when there are tiny errors in any of these inputs. In the horserace with four asset classes, there are just fourteen parameters to estimate and even with such a low number mean-variance does